INCH-POUND

A-A-59378A
November 13, 2001
SUPERSEDING
A-A-59378
December 4, 1998

COMMERCIAL ITEM DESCRIPTION

STOVE, EXPEDITIONARY, MILITARY FUEL

The General Services Administration has authorized the use of this commercial item description for all Federal Agencies.

1. SCOPE

1.1 Scope. This commercial item description (CID) covers a compact, lightweight, expeditionary style stove capable of burning logistically available military type liquid fuels.

1.2 Intended use. The expeditionary stove is intended to be used by personnel engaged in remote locations involving combat operations, or for use in humanitarian relief efforts.

2. SALIENT CHARACTERISTICS

2.1 Functional requirements. The stove shall be a single burner, open flame, with heat control varying from low to high flame. The burner shall be remotely connected by means of a flexible fuel line and a separate pump and fuel storage bottle supplied with the stove. The stove shall be collapsible for placement in a carrying container. The stove shall have minimal parts or components to facilitate easier setup and repacking, and to reduce the risk of misplacing loose parts.

2.2 Weight. The weight of the fuel bottle/burner type stove combination shall not exceed 25 ounces (oz.). The weight includes the burner, pump, empty fuel bottle, carrying container, and accessories.

2.3 Materials. Materials used in the construction of the stove shall be heat and corrosion resistant and shall provide suitable strength to conform to the maximum weight objective (see 2.2), and the performance objective stated herein.

2.3.1 Cookware support. A stable platform having a minimum support diameter of five inches shall be provided as an integral part of the stove.

2.3.2 Stove stability. The stove shall provide stability and support when it is set up on uneven or rough surfaces. Detachable or folding legs shall be provided to stabilize the stove on rough or uneven terrain.

2.3.3 Burner windshield. A means shall be provided to protect or shield the priming and burner flame from the wind such that proper preheat temperatures can be reached in order to ignite the fuel and maintain a uniform shape and flame color without wind distortion.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Soldier and Biological Chemical Command, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:12Z
Check the source to verify that this is the current version before use.

2.4 Fuel. The stove shall be capable of burning the following military fuels available in a battlefield environment:

- JP8, conforming to NATO Code No. F-34 of MIL-T-83133
- Diesel fuel, grade No. 1-D of A-A-52557

2.5 Burner System. The burner shall be equipped with all accessories and fuel jets required to burn the fuels specified in 2.4. The fuel jet shall be permanently identified to distinguish it for the particular fuel it is to burn. The burner shall be capable of operating from sea level to 16,000 feet above sea level without diminished flame performance, and at temperatures ranging from -25°F to 125°F. The burner shall have a minimum heat output of 9500 BTU per hour. The stove shall safely burn the fuel without flareup or flashing of the flame. All components of the stove, including seals and O-rings, shall be capable of withstanding operational temperatures of the stove.

2.5.1 Priming. The burner shall be self-priming using fuel from within the stove. The fuel shall be distributed around the base of the burner to heat it up for the fuel vaporization and ignition stage. A priming wick or priming pad shall be provided to support this preheat stage. Preheat pastes shall not be used to prime the stove. The burner shall be readily and safely ignitable without dangerous flareup.

2.5.2 Fuel control. The burner shall have a fuel control to adjust the flame over a heating range from low to high heat. Flame adjustment shall control the rate and volume of fuel flow into the burner such that all the fuel is vaporized without allowing the fuel to surge into the burner causing flooding or flareup.

2.6 Fuel bottles. Fuel bottles shall be designed to withstand a minimum pressure of 300 psi. The cap or pump shall screw onto the neck of the bottle to maintain a pressure tight, leakproof seal.

2.6.1 Bottle capacity. Fuel bottle capacity shall be standard 0.6 litre (nominal 20 fluid oz.). As an option, a larger 1.0 litre (nominal 34 fluid oz.) size may be provided if specified by the purchaser. (See 6.1)

2.7 Pump. A pump device for connection (screw threads) to the fuel tank or bottle shall be provided. The pump shall create adequate pressure to pump the fuel and maintain flow to the burner unit. The fuel line connection from the pump connected at the fuel bottle shall be of a metal braided, flexible type to prevent undue stress and leaks at the fittings. All connection devices shall be designed for a leak tight fit at operational pressures.

2.8 Protective container. The stove shall be placed in a reusable container rugged enough for use in combat environments. The container shall be large enough to hold the stove, pump, and accessories. The fuel bottle shall be carried separately.

2.8.1 Spare parts. A spare parts kit shall be provided for the stove including all necessary standard components (such as O-rings, pump cup seal, filters, jets, retainers, needles, etc.), and tools to service the stove in the field (see 6.1).

2.9 Operating instructions. Operating instructions shall be provided for fueling, operating, and maintaining the stove. The instructions shall contain all warnings and safety recommendations.

2.10 Maintenance. The stove shall be maintainable and cleanable in the field. The burner shall be equipped with cleaning kit or a built-in cleaning needle for maintaining the fuel jet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:12Z
Check the source to verify that this is the current version before use.

2.11 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of IEEE/ASTM SI 10, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

3. REGULATORY REQUIREMENTS

3.1 Contractor-recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS

4.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, shall conform to the producer’s own drawings, specifications, standards, and quality assurance practices, and be the same product that has been sold in the commercial marketplace, or is the same product that has successfully been delivered to the Government on a previous contract or purchase order. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.2 Market acceptability (MA) criteria. The following market acceptability criteria is necessary to document the quality of the product to be provided under this CID:

a. The company producing the item must have been producing a product meeting the requirements of this CID for at least 1 year.
b. The company must have sold 1000 stoves meeting this CID, in the commercial or government marketplace over the past year.

4.3 Warranty. Unless otherwise specified in the contract, the manufacturer s standard commercial warranty shall apply.

5. PACKAGING

5.1 Packaging. Preservation, packing, labeling, and marking requirements shall be as specified in the contract or purchase order (see 6.1).

6. NOTES

6.1 Ordering Data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents (if applicable).

a. Title, number, and date of this document.
b. Bottle capacity required (see 2.6.1).
c. Spare parts kit (see 2.8.1).
d. Packaging requirements (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:12Z
Check the source to verify that this is the current version before use.

6.2 Sources of documents.

6.2.1 Sources for non-government association documents.

IEEE/ASTM STANDARD

SI 10 - Standard for Use of the International System of Units (SI): The Modern Metric System

(Application for copies of the IEEE/ASTM standard should be addressed to the Institute of Electrical and Electronics Engineers, Inc., 345 East 47th Street, New York, NY 10017, or the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428)

6.2.2 Source for Government documents. Copies of military and federal documents are available from:

Defense Automated Printing Service
Building 4D (DPM-DODSSP)
700 Robbins Avenue
Philadelphia, PA 01911-5094

6.3 Sources of supply. A manufacturer whose product is known to meet the requirements of this document is listed below; however, competition is not limited to this company.

MSR
3800 First Avenue South
P.O. Box 24547
Seattle, WA 98124
(206) 624-8573 V

OPTIMUS USA
The Brunton Company
620 East Monroe
Riverton, WY 82501
(307) 856-6559

MILITARY INTERESTS:

Custodians
Army - GL
Navy - SA
Air Force - 99

Review Activities
Army - MD1, QM1
Air Force - 35, 84
Navy – MC
DLA - IS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

ARMY - GL

(Project 7310-0933)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:12Z
Check the source to verify that this is the current version before use.